

Heisenberg Exchange Metrology to Advance MRAM Performance

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A critical MRAM performance metric is the energy barrier between the memory states. The reversal process down to $\cong 30$ nm device diameter is governed by domain wall mediated switching and the energy barrier consequently depends on the domain wall energy. The relevant materials parameters are the saturation magnetization, perpendicular anisotropy and the Heisenberg exchange stiffness. So far, the Heisenberg exchange stiffness has been challenging to determine for ultrathin films relevant for MRAM free layers. Here, we are measuring the switching probability of MRAM devices and determining the exchange stiffness from fits to the data using a modified droplet model. The results are compared with Brillouin Light Scattering measurements of the spin-wave dispersion of unpatterned identical film stacks. Both measurement approaches show the same trend for the Heisenberg exchange stiffness with Tungsten insertion layer thickness but the unpatterned film measurements show consistently smaller values than the device-level measurements.

Index Terms— MRAM, Heisenberg exchange stiffness, spintronics, switching field distribution

I. INTRODUCTION

MAGNETIC RANDOM ACCESS MEMORY (MRAM) is currently replacing embedded FLASH memories in microcontrollers and is being considered as a candidate for last level cache applications. Many efforts are on-going to further advance MRAM performance including new free layer materials and enhancing CoFeB-based free layers through heterostructuring, such as by insertion of Ta, W or Mo layers or adjacent ferromagnetic layers. Currently, full devices are being fabricated to evaluate any performance increase through such free layer engineering. To address this use of time and resources, we are developing measurements, to characterize all the critical magnetic materials parameters of candidate free layers with an emphasis on film-level measurement of the Heisenberg exchange stiffness, with the ultimate goal to improve models to be predictive of the device performance. An important performance metric for MRAM is for example the energy barrier between the two memory states, which determines the thermal stability and thus impacts data retention.

Determination of the Heisenberg exchange for common free layer thicknesses is challenging. Measurements of perpendicular standing spin-wave modes with ferromagnetic resonance spectroscopy cannot be done on such thin films and accurate temperature-depending magnetometry requires knowledge of the spin-wave gap, which especially impacts the analysis of ultrathin films with low Heisenberg exchange stiffness A_{ex} . Here, we are using two separate approaches: 1) film-level Brillouin Light Scattering spectroscopy (BLS) to measure the wavevector dependence of the spin-waves in the free layer and fit the data to the spin-wave dispersion [1] and 2) measurement of the switching-field distribution (SFD) of MRAM devices and analysis with a model for domain wall driven switching that determines the exchange stiffness[2].

II. EXPERIMENT

Devices and corresponding blanket films with five different free layers have been prepared. The stacks consist of a seed layer, a CoFeB-based reference layer followed by an MgO barrier and the free layer with a capping layer. The CoFeB free layers have

single and double W insert layers with thicknesses t_W between 0.015 nm and 0.320 nm. The total free layer thicknesses t_{FL} , including the W layer, range from 1.205 nm to 2.21 nm, see Table I. The films were deposited by magnetron sputtering and annealed at 335°C for 1 hour. The MgO barriers were rf-sputtered from an MgO target. The resistance-area product ranges from $RA \cong 9.2 \Omega\mu m^2$ to $11.5 \Omega\mu m^2$ and the tunneling magnetoresistance ratio ranges from $\cong 85\%$ to 145% . Both were measured on annealed films by current-in-plane tunneling at room temperature. Table I shows the results for the damping α , the inhomogenous linewidth broadening ΔH_0 and the effective magnetization M_{eff} measured by ferromagnetic resonance spectroscopy.

TABLE I
MAGNETIC PROPERTIES FROM FERROMAGNETIC RESONANCE

t_{FL} (nm)	t_W (nm)	# of inserts	$\mu_0\Delta H_0$ (mT)	α	μ_0M_{eff} (T)
1.205	0.015	1	25.2 ± 0.2	0.0032 ± 0.0002	-0.303 ± 0.001
1.3	0.04	1	22.7 ± 0.2	0.0036 ± 0.0001	-0.299 ± 0.001
2.12	0.23	2	23.9 ± 0.3	0.0090 ± 0.0003	-0.381 ± 0.003
2.12	0.23	2	18.4 ± 0.3	0.0086 ± 0.0002	-0.309 ± 0.004
2.21	0.32	1	23.3 ± 1.2	0.0156 ± 0.0002	-0.488 ± 0.007

The blanket films were measured by BLS in the Damon-Eshbach (DE) and the backward volume (BV) geometry, where the external applied field is applied perpendicular or parallel to the plane of incidence of the 532 nm laser beam, respectively. The applied magnetic field is sufficiently large so that the samples are saturated in-plane. The spin-wave frequency is measured in both geometries over a range of wavevectors and the data are simultaneously fit to the dispersion relations for both the DE and BV geometries to obtain A_{ex} . See Figure 1 for an example measurement and fit for the 1.205 nm thick film. The effective magnetization M_{eff} and the g factor are both required for the fits and are determined from in-plane ferromagnetic resonance spectroscopy.

The SFD measurements were carried out on MTJ devices from 50 nm to 1000 nm nominal diameter, but the extracted parameters shown are for diameters < 120 nm. The switching probability data was fit to a Neel-Brown relaxation model, which allows for the determination of A_{ex} :

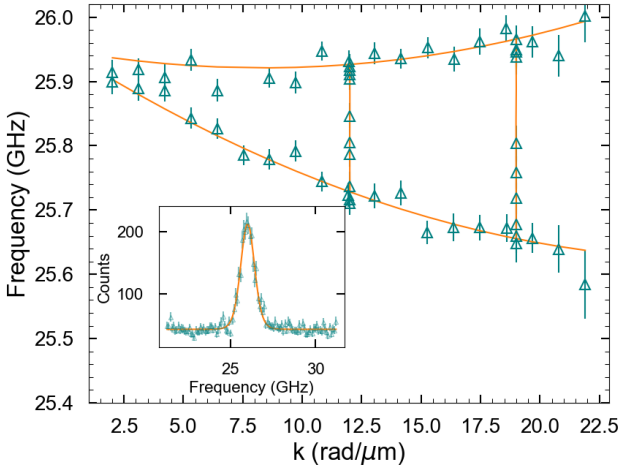


Fig. 2: Spin-wave dispersion data and fits for the 1.205 nm thick free layer sample. The upper branch is the DE and the lower the BV. The vertical lines are measurements at a fixed wavevector but with spin-waves propagating at different angles with respect to the applied field. The inset shows a fit of the Anti-Stokes peak.

$$P(H) = 1 - \exp \left[-f_0 t \cdot \exp \left[-\frac{E_b(H)}{k_B T} \right] \right],$$

where f_0 is the attempt frequency, t the dwell time at field H , and $E_b(H)$ the energy barrier. $E_b(H)$ is calculated assuming a droplet model, where a finite width for the domain wall is being considered. The results for A_{ex} obtained from continuous films and from SFD measurements are shown in Fig. 2. For both methods, the exchange values are smaller than what is found for thick CoFeB films and decrease with Tungsten thickness.

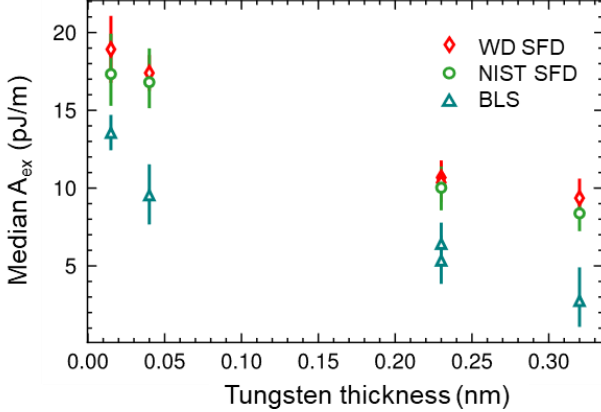


Figure 1: Comparison of the Heisenberg exchange stiffness obtained from SFD and BLS measurements.

III. DISCUSSION AND CONCLUSION

In conclusion, blanket film and device measurements both show the same decreasing trend for the Heisenberg exchange stiffness versus the Tungsten insertion layer thickness, but the values obtained with BLS are consistently lower than the device level measurements. Nevertheless, based on the current results, it appears to be already possible to predict, based on continuous film measurements, the relative magnitude of the energy barrier

for different materials. Once the origin of the difference in A_{ex} for the two different approaches is understood, it is possible that the magnitude of the energy barrier for the modified droplet model can be quantified based only on blanket film measurements. The required parameters are A_{ex} , the perpendicular anisotropy and the saturation magnetization, where the latter two can be determined from FMR and magnetometry. Correlation with other important device metrics is an emerging area of interest. This has the potential to accelerate material screening for MRAM free layers.

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